

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. **Edge AI for Intelligent 5G Network Optimization** (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Summary

Massive devices connected through fifth-generation (5G) networks constitute a ubiquitous Internet of Things (IoT), providing diverse service applications in a smart city. A robust network topology structure against cyber-attacks is the foundation of highly reliable service quality, especially in next-generation networks or beyond 5G (B5G) networks. Existing methods apply neural networks with deep reinforcement learning methods to advance the network topology. However, the reduction of unique hardware resource constraints and the application of edge intelligent networking capability of terminal nodes are emerging challenges for robustness optimization of IoT with 5G and B5G networks. To address these problems, we design a distributed

learning framework utilizing edge intelligence, improving smart terminal nodes' networking capability, which is deployed on ordinary computers instead of specialized hardware such as GPUs. The proposed framework leveraging multi-core CPU and intelligent edge methods decreases the training time and economic cost and takes full advantage of computer resources. Furthermore, the best performing framework considers the distributed communication model of edge computing and optimizes the network topology by taking advantage of smart terminal nodes' contributions. We show that the framework succeeds in various topologies and outperforms compared with other state-of-art algorithms in improving the robustness for IoT topology in smart cities.

The robust network topology provide rich application services for the Internet of Things (IoT) [1] with the emerging technologies for fifth-generation (5G) and beyond 5G (B5G) networks. Nowadays, a stable adaptive robust topology is particularly important due to the complexity of application scenarios and the variability of network service requirements, such as intelligent agriculture [2], smart city [3], smart ocean [4], intelligent transportation [5], etc. Benefit for the high speed and bandwidth of 5G and B5G, massive smart sensor devices can be connected to form a highly heterogeneous complex IoT. Failure of some nodes causes the global network's chain collapse effect, which makes the global network unable to communicate. Therefore, designing reliable network topologies for 5G and B5G networks are drawn much attention from researchers. A variety of solutions have been proposed to improve network topology's robustness [6–8]. These algorithms share a common idea: the best optimum topology is searched by the evolution framework and takes more computing resources and time. Robustness optimization of network topology represents improving the resistance to cyber-attacks, including self-failure, invasion attacks, disaster attacks, etc. Many existing algorithms evaluate performance with several types of attacks. Some proposed learning frameworks [3] utilize centralized architectures to optimize the networking for IoT in smart cities, which can dynamically and intelligently adjust the IoT topology to maintain high communication capability. However, these frameworks rely too heavily on specialized hardware such as GPUs and ignore smart terminal nodes' contributions to the IoT topology in a smart city. With the rise of edge computing technology with 5G and B5G [9], smart terminal nodes have an important

influence on the whole network robustness, which is considered to optimize the IoT network topology

which provides better data analysis and communication capacity services for applications in 5G and B5G networks.

In this article, we design a very different paradigm for the robustness optimization of network topology for IoT, referred to as edge computing communication methods. Instead of a single centralized neural network training model, we asynchronously execute distributed multiple models parallel with multiple instances of the network topology, which includes a global learning model and several edge local learning models [10]. Each learning instance is like an edge learning node, responsible for optimizing the local model and then communicating with the global model. The paradigm also decorates the models' data into a stationary process because each model experiences various states at each training step. Each model runs a deep reinforcement learning framework which is input different IoT topology data. This strategy enables more kinds of samples on the model with more edge nodes, which learn practical optimization approaches, accelerate the convergence, and improve performance.

Our proposed parallel distributed networking robustness optimization mechanism also offers practical benefits. Whereas previous approaches to machine learning [3] rely heavily on specialized hardware such as GPUs, our mechanism executes on a single machine with a standard multi-core CPU [10]. Besides, we run a distributed communication mechanism according to edge computing methods. Furthermore, we consider a general attack method to evaluate the performance of the algorithm [3, 8, 11], namely Degree Centrality. The best of the proposed method masters several state-of-art existing solutions. Nowadays, 5G technology can be applied to many scenarios of smart cities, i.e., intelligent transportation, smart drones, monitoring system, smart office, etc., as shown in Fig. 1. The server stores the topology of a smart

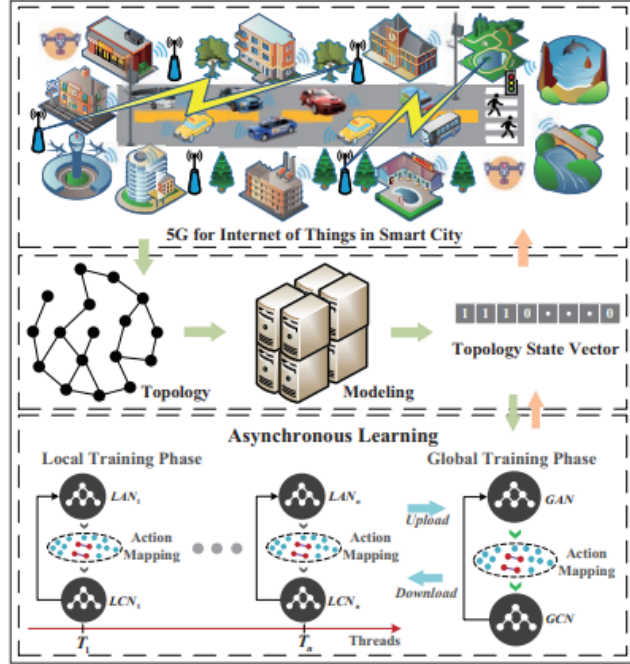


FIGURE 1. In a smart city, the topology of 5G and B5G for IoT can be modeled by servers and then converted to state vector. We input the topology state vector to the asynchronous learning model. In the local training phase, each thread T executes an exclusive deep reinforcement learning model. Individual topology optimization policies are obtained through the local actor-network LAN, action mapping operator, and local critic network LCN. Threads share the global learning phase, and global actor-network GAN and global critic network GCN update their parameters asynchronously through pulling each local training model.